Diploma in Computer Science & Information Technology

SBTE, Bihar — Semester IV

Computer Troubleshooting & Maintenance

Course Code: 2418405

Unit-wise Detailed Notes

UNIT 1: Internal Components of the PC

1.1 Types of Computers

- **Desktop Computer:** Stationary system unit with separate monitor, keyboard, mouse. High performance, upgradeable.
- **Laptop/Notebook:** Portable, all-in-one design with integrated display, keyboard, battery.
- **Tablet:** Touch-screen portable device with no physical keyboard. Uses mobile OS.
- **Mainframe:** Large, powerful computers for bulk data processing. Used by banks, government.
- **Supercomputer:** Most powerful; used for complex scientific simulations, weather forecasting.

1.2 Motherboard

The motherboard (mainboard) is the main printed circuit board that connects all computer components. It provides electrical connections through buses and slots.

Key Motherboard Components

- **CPU Socket:** Physical connector for the processor.
- **RAM Slots (DIMM slots):** Slots for memory modules.
- **Chipset (North Bridge + South Bridge):** Controls communication between CPU, RAM, and peripheral devices.
- **Expansion Slots:** PCIe slots for graphics cards, PCIe x1 for other cards.
- **SATA Connectors:** For connecting hard drives and optical drives.
- **BIOS/UEFI Chip:** Stores firmware for booting.
- **Power Connector:** 24-pin ATX connector from power supply.

1.3 Processor (CPU)

The CPU is the "brain" of the computer. It executes instructions and processes data. Key specifications: Clock speed (GHz), Number of cores, Cache size (L1, L2, L3), Thermal Design Power (TDP).

Multi-Core Processors

A multi-core processor has multiple processing cores on a single chip. Dual-core: 2 cores; Quad-core: 4 cores; etc. Each core can execute instructions independently, improving parallel processing and multitasking performance.

1.4 BIOS (Basic Input/Output System)

BIOS is firmware stored on a ROM chip on the motherboard. It runs when the computer powers on (Power-On Self Test — POST), initializes hardware, and loads the operating system bootloader.

BIOS Settings

- Date and Time configuration
- Boot device priority (which drive to boot from first)
- Enable/disable hardware components
- Fan speed and temperature thresholds
- BIOS password for security

UEFI (Unified Extensible Firmware Interface) is the modern replacement for traditional BIOS — supports larger drives (GPT), faster boot, graphical interface.

1.5 System Memory (RAM)

RAM (Random Access Memory) is volatile main memory — loses data when power is off. Temporarily stores programs and data currently in use by the CPU.

Memory Module Types

- DRAM (Dynamic RAM): Must be constantly refreshed. Most common main memory type.
- SDRAM: Synchronized to the CPU clock for higher performance.
- DDR SDRAM: Double Data Rate — transfers data on both rising and falling clock edges. DDR4 and DDR5 are current standards.
- Memory form factors: SIMM (72-pin), DIMM (168/184/240-pin for desktops), SO-DIMM (for laptops).

1.6 Power Supply (SMPS)

The Switched-Mode Power Supply (SMPS) converts AC mains power to DC voltages (+12V, +5V, +3.3V, -12V) required by computer components.

UPS (Uninterruptible Power Supply): Provides battery backup during power outages. Offline UPS: switches to battery after power failure. Online UPS: always running on battery (best protection).

Troubleshooting Power Supply

Use a multimeter to test output voltages. Dead computer with no POST beep often indicates PSU failure. Check all cable connections. Power supply failure symptoms: random shutdowns, burning smell, no power at all.

UNIT 2: Input Device, Output Device, and Storage Devices

2.1 Keyboard and Mouse

Keyboard operation: Each key press generates a unique scan code sent to the CPU via keyboard controller. Types: Mechanical (tactile feedback), Membrane (quiet), Ergonomic. Interfaces: PS/2 (round connector), USB (most common), Bluetooth (wireless).

Mouse: Tracks movement and translates to cursor position. Types: Mechanical (ball), Optical (LED/laser), Wireless. Interfaces: PS/2, USB, Bluetooth. Mouse common problems: cursor jitter (clean sensor), double-click issues (switch replacement).

2.2 Scanners

Scanners convert physical documents/images into digital data. Types: Flatbed scanner (glass platen), Sheet-fed scanner, Handheld scanner, 3D scanner. Image quality measured in DPI (Dots Per Inch) — higher DPI = better resolution.

2.3 Printers

Inkjet Printer

Sprays tiny droplets of ink onto paper. Components: Print head (nozzles), Ink cartridges, Paper feed mechanism, Carriage rod. Problems: clogged nozzles (run cleaning cycle), poor print quality (replace cartridge), paper jams (remove obstructions carefully).

LaserJet Printer

Uses electrostatic process: Drum charged, laser writes image, toner attracted, fused to paper by heat. Components: Laser unit, Photosensitive drum, Toner cartridge, Fuser unit, Transfer belt/roller. Faster and cheaper per page than inkjet for high-volume printing.

2.4 Monitor and Display Technologies

CRT (Cathode Ray Tube): Old technology, heavy, bulky. LCD (Liquid Crystal Display): Flat, energy-efficient, most common. LED: LCD with LED backlight — thinner, brighter. OLED: Each pixel emits its own light — perfect blacks, vibrant colors. Plasma: Used in large TVs, now largely replaced by LED/OLED.

VGA, DVI, HDMI, DisplayPort are common video connector standards. Graphics Cards: GPU processes graphics data. AGP (old), PCIe x16 (current standard for discrete GPUs).

2.5 Hard Disk Drive (HDD)

HDD stores data magnetically on spinning platters coated with magnetic material. Read/write heads access data as platters spin at 5400-7200+ RPM.

Disk Geometry

- Track: Concentric circles on a platter surface.
- Sector: Smallest addressable unit (typically 512 bytes or 4096 bytes).
- Cylinder: All tracks at the same position across all platters.
- Cluster: Group of sectors — the file allocation unit.

HDD Interfaces

- SATA (Serial ATA): Current standard. SATA III up to 6 Gbps.
- SCSI: High performance for servers.
- USB/IEEE 1394: For external drives.

Solid State Drive (SSD)

SSDs use flash memory chips — no moving parts. Much faster than HDD, shock-resistant, silent. Interfaces: SATA, M.2 (NVMe PCIe for highest speed). Limited write cycles compared to HDD.

UNIT 3: Software, Malware, Security, and Drivers

3.1 Operating System

The OS manages all hardware and software resources and provides a user interface. Installation steps: Boot from installation media, partition disk, select file system (NTFS, ext4), install files, configure settings, install drivers.

OS Troubleshooting Scenarios

- "Missing Operating System" / "BOOTMGR is missing": Boot from installation media, use Startup Repair or bootrec /fixmbr command.
- Blue Screen of Death (BSOD) / Kernel Panic: Note error code, check recent driver/hardware changes, run memory test, check disk for errors.
- Slow startup: Disable unnecessary startup programs, scan for malware, check SSD health.
- System crashes: Run sfc /scannow (Windows), check RAM, update drivers.

3.2 Malware and Security

Malware (Malicious Software) is any software designed to harm, exploit, or gain unauthorized access. Types: Virus (attaches to programs), Worm (self-replicating over networks), Trojan (disguised as legitimate software), Ransomware (encrypts files for ransom), Spyware (collects information secretly), Adware (displays unwanted ads), Rootkit (hides deep in system).

Detection and Removal

- Boot into Safe Mode to prevent malware from loading.
- Run full scan with updated antivirus (Windows Defender, Malwarebytes, etc.).
- Use bootable antivirus media for persistent infections.
- After removal: reset passwords, check startup programs, update all software.

3.3 Device Drivers

A device driver is software that allows the OS to communicate with hardware devices. Without drivers, hardware cannot function. Types: Kernel-mode drivers (high privilege), User-mode drivers, Plug-and-Play drivers (auto-detected and installed).

Driver Installation and Troubleshooting

Install from manufacturer's website or Windows Update. Driver conflicts: Two drivers conflict for same resource. Solution: Update or roll back driver in Device Manager. Yellow exclamation mark in Device Manager indicates driver problem.

Driver signing: Modern OS (64-bit Windows) requires digitally signed drivers. Use proper official drivers to avoid instability.

UNIT 4: Network and Internet Connectivity

4.1 Networking Devices

Switch (Layer 2 Device)

A switch connects multiple devices in a LAN and forwards frames based on MAC addresses. It learns MAC addresses by inspecting source addresses of incoming frames (MAC address table). Unlike a hub, it sends data only to the intended recipient (reduces collisions). Uplink port: Connects switch to another switch or router.

Router (Layer 3 Device)

A router connects different networks (LAN to WAN/Internet) and forwards packets based on IP addresses using a routing table. Basic router configuration: Set IP address, subnet mask, default gateway, DHCP settings, NAT (Network Address Translation), SSID and security for wireless.

4.2 Internet Connectivity Types

- **Broadband:** High-speed internet always-on connection. Includes DSL (phone lines), Cable (coaxial), Fiber optic (fastest).
- **Leased Line:** Dedicated, symmetric connection between two points. Fixed bandwidth, guaranteed. Used by businesses.
- **Wireless (Wi-Fi):** IEEE 802.11 standard. 802.11n, 802.11ac, 802.11ax (Wi-Fi 6) are current standards.
- **Mobile Data (4G/5G):** Cellular network internet access.

4.3 Firewalls and Security

A firewall monitors and controls incoming and outgoing network traffic based on predefined security rules. It forms a barrier between trusted internal network and untrusted external network.

Types of Firewalls

- **Packet Filtering Firewall:** Examines each packet's header (IP, port, protocol) and allows or blocks based on rules. Fast but no state tracking.
- **Stateful Inspection Firewall:** Tracks active connections; understands state of network sessions.
- **Application-Layer Firewall (Proxy):** Operates at layer 7; can inspect application content.
- **Hardware Firewall:** Dedicated device (router with built-in firewall). Protects entire network.
- **Software Firewall:** Installed on individual computers. Windows Defender Firewall, iptables (Linux).

VPN (Virtual Private Network)

A VPN creates an encrypted tunnel over the internet, allowing secure remote access to a private network. Protects data from interception on public networks.

UNIT 5: Data Recovery and System Performance Optimization

5.1 Causes of Data Loss

- **Human Error:** Accidental deletion, formatting wrong drive, overwriting files.
- **Hardware Failure:** HDD head crash, SSD failure, power surge damage.
- **Software Corruption:** File system corruption, failed OS update, application crashes.
- **Viruses and Malware:** Ransomware encrypts files; other malware deletes/corrupts data.
- **Power Outages:** Sudden power loss during write operations can corrupt file system.
- **Natural Disasters:** Fire, flood, lightning strikes damaging hardware.

5.2 Data Recovery Methods

When data is "deleted," the OS typically just marks the space as available without erasing the actual data. Recovery is possible until that space is overwritten.

- **File Restore:** Recover from Recycle Bin (Windows) or Trash (macOS/Linux) if not permanently deleted.

- **Backup Restore:** Restore from previously created backup (most reliable method).
- **Data Recovery Software:** Recuva, TestDisk, PhotoRec, R-Studio. Scan storage device for recoverable files. Best results when drive has not been heavily used after data loss.
- **Professional Recovery:** For physical damage (head crash) — requires cleanroom environment. Expensive but sometimes the only option.

5.3 Data Backup Strategies

- **Full Backup:** Complete copy of all selected data. Slowest to create, fastest to restore.
- **Incremental Backup:** Only backs up data changed since the last backup (full or incremental). Fast to create, slower to restore (need full + all incrementals).
- **Differential Backup:** Backs up all data changed since the last FULL backup. Slower to create than incremental, faster to restore (need only full + latest differential).
- **Cloud Backup:** Automatic off-site backup to cloud services. Protects against local disasters.

Note: Follow the 3-2-1 backup rule: 3 copies of data, on 2 different media types, with 1 off-site copy.

5.4 System Performance Optimization

Software Optimization

- Manage Startup Programs: Disable unnecessary programs from auto-starting (Task Manager > Startup in Windows).
- Registry Cleaning: Remove invalid/outdated registry entries using tools like CCleaner. (Use carefully — backup registry first.)
- Disk Cleanup: Remove temporary files, downloaded program files, and Recycle Bin contents.
- Defragmentation: Reorganize fragmented files on HDD for faster access. (Not needed/useful for SSDs.)
- Update Drivers and OS: Keep operating system and drivers updated for security and performance.

Hardware Optimization

- Overheating Solutions: Clean dust from CPU heatsink and fans. Replace dried thermal paste between CPU and heatsink. Improve case airflow.
- RAM Upgrade: Adding more RAM reduces paging (virtual memory use) and improves multitasking.
- SSD Upgrade: Replacing HDD with SSD dramatically improves boot time and application loading.
- GPU Upgrade: Better graphics card for improved gaming and GPU-accelerated tasks.